

Slides 6.5: Sustainable Food Methods and the Food Systems Quiz

Slide outline for Lesson 6.5. Twenty slides across two days: Day 1 slides 1 to 11, Day 2 slides 12 to 20. Cover the wall paths before slide 20. *[Sal: collect the packages for a week before you teach this. Ask students to bring in the worst package they can find; they compete to win that.]*

DAY 1

Slide 1: Sustainable Food Methods

- Unit 6, Lesson 6.5, Day 1
- Topic 6.1 Where Food Comes From
- Today: you plant something and you keep it alive until the end of this unit Image: a row of paper cups with seedlings on a classroom windowsill.

Slide 2: Do now

- Something is sustainable if you can keep doing it forever without running out
- Name one thing you do that is sustainable
- Name one thing that is not
- They can be small Notes: Take four pairs of answers.

Slide 3: This is trash from Tuesday

- A scallion root end
- It will be a plant by next week
- You each get one today Image: a scallion root end in a clear cup of water with new green growth. Notes: This is the hook. Hold up the real one.

Slide 4: Sustainability, the definition

- The state's own words: define "sustainability" as it applies to resource use
- In plain words: a way of using something you could keep doing without running out or wrecking the place it comes from Notes: Students write it in their own words.

Slide 5: Crop rotation

- Plant a different crop in the same field each year instead of the same one over and over
- The soil keeps its nutrients
- Pests and diseases that live on one crop lose their host Image: a four-square diagram of one field with a different crop each year, arrows in a circle.

Slide 6: Drip irrigation

- Hoses along the rows drip water right at the roots
- Instead of spraying the whole field into the air
- Far less evaporates or runs off
- Same crop, much less water Image: a close-up of drip tape beside a row of tomato plants.

Slide 7: Integrated pest management

- Watch for pests and count them
- Use the smallest thing that works: a resistant seed, a helpful insect, a trap, a barrier
- Spray last, not first

- Costs less and keeps beneficial insects alive Image: a sticky trap card in a field with a farmer's hand holding a clipboard.

Slide 8: Plant it

- Scoop the mix
- Poke a hole, seed or root end in, cover
- Water once
- Label with your name and today's date
- Then write the care plan: how much, how often, how much light, who waters when you are absent Notes: Run it as a line. Wash hands after. Assign the watering chart, four students per week, all the way to the end of Unit 6.

Slide 9: Compost sort

- Two piles: goes in, does not go in
- A reason for every card in the "does not" pile
- "It is gross" is not a reason Notes: Ten cards per group of four. Let them argue about the paper towel and the slimy lettuce.

Slide 10: The compost rule

- Things that used to be a plant break down
- Things that came from an animal, or have oil or grease on them, draw animals and turn sour instead of breaking down
- Plastic never breaks down, and stickers are the number one contaminant in commercial compost Notes: Reveal the ten answers with the why, not just the answer.

Slide 11: One cucumber, two ways

- Wrapped in plastic: stays fresh about three times as long
- Loose: no plastic, and more of them get thrown out
- Which waste is worse? Image: two cucumbers side by side, one shrink wrapped, one bare. Notes: This sets up Day 2. Hand out the three packages per group and have them circle the one they will redesign. Exit card: "Sustainable means . ***One thing that makes food production more sustainable is .***"

DAY 2

Slide 12: Redesign the Package

- Unit 6, Lesson 6.5, Day 2
- Build it, test it, pitch it
- Then the Food Systems Quiz in the last fifteen minutes Image: a single-serve yogurt cup inside a cardboard sleeve inside a shrink-wrapped four-pack.

Slide 13: Do now

- Look at the three changes on the wall from the waste audit
- Which one actually happened?
- Be honest Notes: Sixty seconds. Honest answers only; no consequence either way.

Slide 14: Four layers to protect one yogurt

- Your job: do it in two
- And prove the yogurt still survives Notes: Hold up the worst package in the pile. This is the hook.

Slide 15: The design process, the state's own eight steps

- Define the problem
- Set criteria and constraints
- Brainstorm
- Select
- Prototype
- Test, optimize, evaluate Notes: Same eight words as Unit 4, Lessons 4.11 to 4.15. Say that out loud so students hear the connection.

Slide 16: Your criteria and your constraints

- **Criteria:** the food still gets protected, and the waste goes down
- **Constraints:** survives a truck and being stacked three high. Lasts two weeks on a shelf. Still says what is inside, with a Nutrition Facts panel Notes: The constraints are given, not chosen. Write them on the board and leave them up.

Slide 17: Build it

- Ten minutes
- Paper, tape, cardboard, markers
- One model per group
- Draw the label on it Notes: Circulate with one question per group: "Drop it. Stack three. Does the food still make it?" A group that removed the structural layer finds out in two seconds.

Slide 18: Justify it

- How many layers did the original have? How many does yours?

- What protects the food now?
- What did you remove, and what was that layer doing?
- What got worse? Notes: Both criteria have to be answered. “What got worse” is not optional; every design gives something up.

Slide 19: Thirty-second pitches

- One sentence: what we cut
- One sentence: how the food is still protected
- One challenge question from the class Notes: Five minutes total. Then collect the models and clear the desks.

Slide 20: Food Systems Quiz

- 12 items, 15 minutes
- Supply chain, local and seasonal, waste math, a food desert scenario, sustainable methods, packaging
- You may use your vocabulary card
- Desks clear, wall covered Notes: Read the directions aloud once. Cover the wall paths before you hand the quizzes out. Collect at the bell. Retakes are allowed once after a review, per the Grading Plan; post the answers at the start of Lesson 6.6.